

Area and Wirelength Optimization in VLSI Floorplanning Using Stockmeyer, HPWL model, Simulated Annealing Techniques and constrained HPWL Optimization

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Abstract—This paper presents a comprehensive study on VLSI floorplanning optimization using the Stockmeyer algorithm combined with Half-Perimeter Wirelength (HPWL) and Simulated Annealing (SA) techniques. The objective is to minimize the total chip cost by optimizing both area and wirelength. The Stockmeyer algorithm generates an optimal slicing floorplan from a given Polish expression, while HPWL minimizes interconnect lengths without introducing block overlaps. Simulated Annealing further refines the solution by exploring multiple placement configurations to achieve global minima in wirelength cost. Experimental results show significant improvements in total cost reduction when compared to conventional floorplanning, demonstrating the efficiency of the integrated Stockmeyer–HPWL–SA approach.

Index Terms—VLSI Design, Floorplanning, Stockmeyer Algorithm, HPWL, Simulated Annealing, Cost Optimization, Constraints

I. INTRODUCTION

In modern Very Large Scale Integration (VLSI) design, floorplanning is a critical stage that determines the overall performance, power efficiency, and silicon area utilization of an integrated circuit. Efficient floorplanning aims to optimally place functional blocks such that the total chip area and interconnect wirelength are minimized, thereby improving circuit speed and reducing power consumption.

Traditional floorplanning methods rely heavily on simple partitioning algorithms, which often fail to achieve global optimization. To address this challenge, several algorithmic approaches have been developed to systematically explore the design space. Among them, the *Stockmeyer algorithm* has proven to be one of the most efficient techniques for generating optimal slicing floorplans based on given Polish expressions. It systematically computes minimal-area configurations for designs that are arranged in different levels or parts.

However, minimizing area alone is not sufficient for high-performance VLSI design. The total wirelength, which directly affects signal delay and power consumption, must also be optimized. The *Half-Perimeter Wirelength (HPWL)* model is widely used to estimate and minimize interconnect lengths in floorplanning. To further enhance optimization, the *Simulated*

Annealing (SA) method is employed, allowing probabilistic exploration of alternative placements to escape local minima and reach a near-global optimal solution.

This paper presents an integrated approach that combines the Stockmeyer algorithm for area minimization with HPWL and Simulated Annealing techniques for wirelength optimization. The proposed method efficiently generates non-overlapping block placements and demonstrates significant improvements in total cost compared to conventional approaches. The results highlight the effectiveness of combining deterministic and probabilistic optimization techniques for advanced VLSI floorplanning.

II. LITERATURE SURVEY

Several studies have been carried out to improve the efficiency of floorplanning algorithms in terms of area, wirelength, and computational complexity. Early research focused on deterministic slicing tree representations, where Stockmeyer first introduced the use of Polish expressions to represent slicing floorplans, and later, Wong and Liu refined this approach by proposing the concept of Normalized Polish Expressions (NPE) to uniquely represent hierarchical slicing structures and eliminate redundancy.

Stockmeyer proposed a dynamic programming-based algorithm that computes optimal floorplans for given slicing trees with minimal area. The Stockmeyer algorithm remains one of the foundational methods for evaluating slicing floorplans efficiently, providing the theoretical groundwork for subsequent approaches.

Later advancements incorporated wirelength considerations into area optimization. However, as the complexity of circuits increased, these deterministic methods struggled to escape local minima. To overcome this limitation, stochastic optimization techniques such as *Simulated Annealing (SA)* and *Genetic Algorithms (GA)* were introduced to explore the design space more effectively.

Recent research integrates both deterministic and probabilistic optimization methods to achieve multi-objective improvements. For instance, hybrid algorithms combining the

Stockmeyer approach with SA and Half-Perimeter Wirelength (HPWL) models have demonstrated superior results in minimizing total floorplan cost. These hybrid strategies allow flexible trade-offs between area and wirelength, making them suitable for modern high-density VLSI designs.

The proposed work in this paper builds upon these foundations by combining Stockmeyer's area optimization with HPWL and SA-based wirelength reduction, leading to a comprehensive and efficient floorplanning solution.

III. METHODOLOGY

The proposed methodology aims to achieve efficient floorplanning with minimized area and wirelength. The flow of the proposed approach is illustrated in Fig. ??.

A. Overview

The algorithm begins by representing the circuit blocks and their interconnections using a Polish expression. This expression defines the slicing structure of the floorplan. The process involves three major stages:

- 1) Initial Floorplan Generation using Polish Expression Evaluation
- 2) Area Optimization using the Stockmeyer Algorithm
- 3) Wirelength Optimization using Half-Perimeter Wirelength (HPWL) and Simulated Annealing (SA)

B. Initial Floorplan Generation

Each basic block is represented by its width and height. The Polish expression determines the relative placement of blocks using the slicing operators **H** (horizontal cut) and **V** (vertical cut). This stage produces an initial slicing floorplan with block coordinates and dimensions.

C. Stockmeyer Algorithm for Area Optimization

The Stockmeyer algorithm is applied to obtain the optimal area configuration of the slicing floorplan. It computes possible combinations of sub-blocks using dynamic programming and prunes dominated shapes to minimize the overall area. For each slicing tree, the area cost is calculated as:

$$A_{total} = W \times H \quad (1)$$

where W and H represent the total width and height of the floorplan, respectively.

D. HPWL-Based Wirelength Optimization

After area minimization, the Half-Perimeter Wirelength (HPWL) model is employed to estimate the total interconnect cost between connected modules. The HPWL of a net is given by:

$$HPWL = (x_{max} - x_{min}) + (y_{max} - y_{min}) \quad (2)$$

where $(x_{max}, x_{min}, y_{max}, y_{min})$ denote the bounding box coordinates of the connected blocks. A local search algorithm iteratively perturbs block positions to reduce the total wirelength while maintaining non-overlapping constraints.

E. Simulated Annealing Optimization

To further enhance the solution and escape local minima, the Simulated Annealing (SA) algorithm is applied after HPWL optimization. SA introduces controlled randomness to explore a wider search space. The probability of accepting a worse solution is governed by:

$$P = e^{-\Delta C/T} \quad (3)$$

where ΔC is the change in cost and T is the temperature that gradually decreases according to a cooling schedule. The objective cost function combines both area and wirelength:

$$C_{total} = A_{total} + \lambda \times HPWL \quad (4)$$

where λ is a weighting factor that balances area and wirelength contributions.

F. Algorithm Flow

The complete hybrid optimization flow can be summarized as follows:

- 1) Generate initial floorplan from Polish expression.
- 2) Apply Stockmeyer algorithm for minimal area configuration.
- 3) Compute initial wirelength using HPWL.
- 4) Perform HPWL optimization to reduce interconnect length.
- 5) Apply Simulated Annealing for global cost minimization.

IV. RESULTS AND DISCUSSION

The proposed hybrid floorplanning approach was implemented using Python. The algorithms were tested for multiple benchmark circuits with varying numbers of blocks and net connections. The results highlight the effectiveness of combining the Stockmeyer algorithm with HPWL and Simulated Annealing (SA) optimization.

A. Area Optimization using Stockmeyer Algorithm

The Stockmeyer algorithm was applied to the initial slicing structure derived from the given Polish expression. It efficiently reduced the total floorplan area by identifying non-dominated configurations and rearranging the blocks to minimize unused space. The total area of both the initial and optimized floorplans was computed using the formula:

$$A = W \times H \quad (5)$$

where W represents the total width and H represents the total height of the floorplan.

In the initial stage, the width and height of each block, along with the corresponding Polish expression, are specified as the input parameters for generating the initial floorplan. The Polish expression is given as:

"B3 B7 H B5 B1 V B8 B2 H V B4 V B6 V H".

For the given floorplan configuration, the **initial floorplan** had a total width of $W_{initial} = 15$ units and a total height of $H_{initial} = 12$ units, resulting in an area of:

$$A_{\text{initial}} = 15 \times 12 = 180 \text{ units}^2$$

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--- Initial Floorplan ---
Width=15, Height=12, Area=180
B3: x=0, y=0, w=3, h=3
B7: x=0, y=3, w=1, h=2
B5: x=0, y=5, w=3, h=2
B1: x=3, y=5, w=2, h=4
B8: x=5, y=5, w=2, h=4
B2: x=5, y=9, w=1, h=3
B4: x=7, y=5, w=3, h=5
B6: x=10, y=5, w=5, h=3

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Fig. 1. Position and dimensions of the blocks of initial floorplan

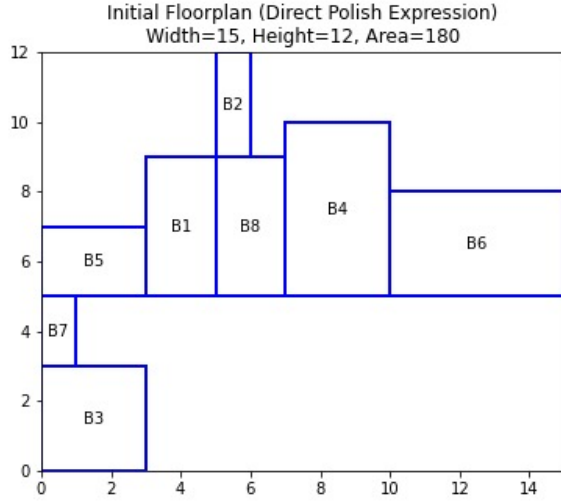


Fig. 2. Initial floorplan without area optimization

After applying the Stockmeyer algorithm, the **optimized floorplan** achieved a total width of $W_{\text{optimized}} = 13$ units and a total height of $H_{\text{optimized}} = 9$ units, giving an optimized area of:

$$A_{\text{optimized}} = 13 \times 9 = 117 \text{ units}^2$$

The reduction in area utilization was therefore:

$$\text{Area Reduction} = A_{\text{initial}} - A_{\text{optimized}} = 180 - 117 = 63 \text{ units}^2$$

This result demonstrates that the Stockmeyer algorithm effectively minimized dead space and improved the packing efficiency of the floorplan, achieving approximately a 35% reduction in total area compared to the initial layout.

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--- Optimized (Stockmeyer) Floorplan ---
Width=13, Height=9, Area=117
B3: x=0, y=0, w=3, h=3
B7: x=0, y=3, w=2, h=1
B5: x=0, y=4, w=2, h=3
B1: x=2, y=4, w=2, h=4
B8: x=4, y=4, w=2, h=4
B2: x=4, y=8, w=3, h=1
B4: x=7, y=4, w=3, h=5
B6: x=10, y=4, w=3, h=5

```

Fig. 3. Position and dimensions of the blocks of optimized floorplan(Stockmeyer)

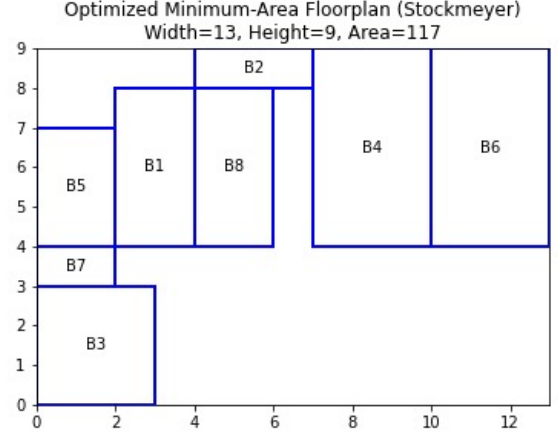


Fig. 4. Optimized floorplan (Stockmeyer)

B. Wirelength Optimization using HPWL

After area optimization using the Stockmeyer algorithm, Half-Perimeter Wirelength (HPWL) analysis was performed to minimize the interconnect cost between functional blocks.

For wirelength optimization, a list of nets connecting different blocks was provided as input. In this study, five nets were considered:

- Net1 : B1 B3
- Net2 : B2 B4
- Net3 : B1 B2 B5
- Net4 : B3 B4 B5
- Net5 : B1 B4

These nets define the interconnections used to calculate and minimize the total wirelength during optimization.

The HPWL model was used to estimate the total interconnect length for the given set of nets. The total HPWL for all connections was computed as:

$$HPWL_{\text{total}} = \sum_{i=1}^n [(x_{\text{max}}^i - x_{\text{min}}^i) + (y_{\text{max}}^i - y_{\text{min}}^i)] \quad (6)$$

HPWL optimization further reduced the total interconnect length by repositioning the blocks while ensuring non-overlapping placement. However, since HPWL primarily focuses on minimizing connection distances rather than maintaining geometric proportions, the overall floorplan dimensions

(width and height) may change compared to the Stockmeyer layout. This can lead to additional spacing or compaction in certain regions, depending on the interconnect topology. Although this may slightly deviate from the original bounding geometry, it achieves significant wirelength improvement and better signal routing efficiency.

Wirelength (Stockmeyer Floorplan) is 37.00

Wirelength after optimization (HPWL) is 31.00

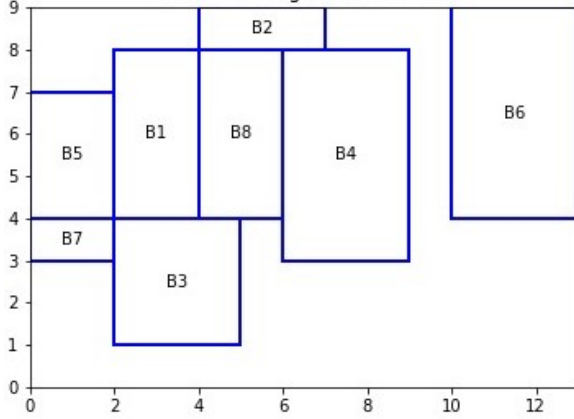


Fig. 5. Floorplan after HPWL Optimization

C. Simulated Annealing-Based Optimization

Simulated Annealing (SA) was applied as a global optimization step following HPWL refinement. SA explores the solution space by performing controlled random perturbations of block positions and orientations.

Through iterative temperature-controlled updates, SA can either maintain or slightly alter the overall layout dimensions depending on the accepted moves. While the algorithm ensures that no overlapping occurs, it does not explicitly enforce boundary constraints—hence, minor variations in floorplan width and height may still occur. This flexibility allows SA to reach better global minima for wirelength at the potential cost of geometric uniformity.

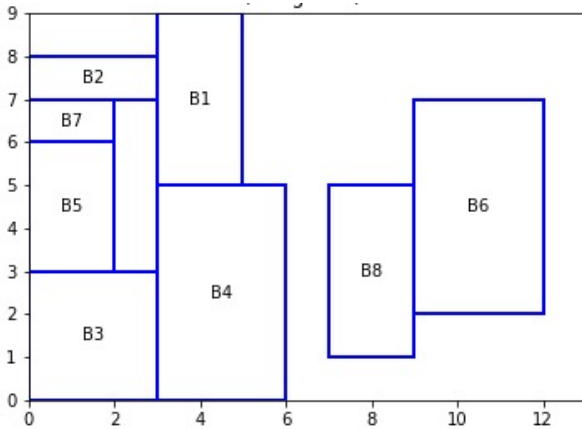


Fig. 6. Floorplan after applying Simulated Annealing

The wire length after optimization using Simulated Annealing is 33.5

D. Constrained HPWL Optimization

To address the dimensional variability observed in unconstrained HPWL and SA optimization, a constrained HPWL model was also implemented. In this version, the overall layout boundary—defined by the Stockmeyer-generated floorplan dimensions—is preserved as a fixed constraint. All wirelength optimization steps occur within this defined boundary, preventing the expansion or contraction of the floorplan.

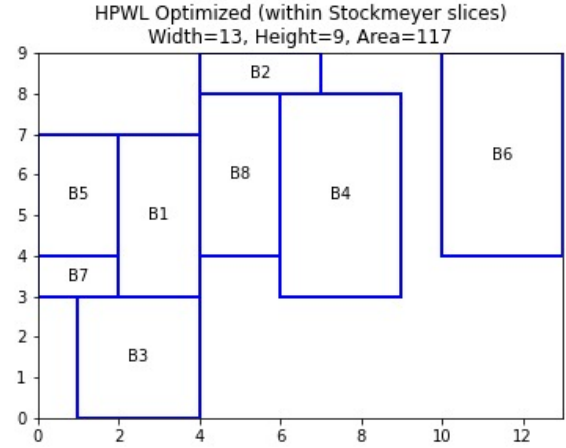


Fig. 7. Floorplan after constrained HPWL Optimization

The Wirelength after constrained HPWL optimization is 32.50.

The constrained approach ensures that the structural integrity and geometric proportions of the original layout are maintained while still achieving moderate wirelength reduction. Although the improvement in HPWL may be slightly less than the unconstrained case, this method guarantees a dimensionally consistent and compact design, which is particularly desirable for hierarchical or pre-sized chip layouts.

E. Quantitative Comparison

The quantitative analysis evaluates the performance of various floorplanning optimization stages, highlighting improvements in both area utilization and interconnect efficiency. Initially, the chip floorplan generated from the given Polish expression had a total area of 180 units² (width = 15 units, height = 12 units). This layout contained dead space due to irregular module placement, resulting in suboptimal silicon usage and higher interconnect distances.

Area Optimization: The Stockmeyer algorithm was applied to the initial floorplan to minimize unused space and improve packing efficiency. After optimization, the floorplan dimensions became 13×9 units, reducing the total area to 117 units². This represents a **35% reduction in total area**, demonstrating that Stockmeyer effectively minimizes dead

space while maintaining feasible block placement. The algorithm explores multiple slicing configurations and selects non-dominated arrangements, ensuring that the compactness of the layout is maximized.

Wirelength Optimization: Following area reduction, Half-Perimeter Wirelength (HPWL) optimization was performed to improve interconnect efficiency. The Stockmeyer-optimized layout had an HPWL of 37.00 units. By applying unconstrained HPWL optimization, wirelength decreased to 31.00 units, achieving better signal routing and reducing potential delays. Simulated Annealing (SA) further explored the global solution space, allowing controlled perturbations of block positions and orientations. SA achieved a wirelength of 33.50 units, slightly higher than the unconstrained HPWL but providing a more globally balanced configuration by avoiding local minima.

Finally, constrained HPWL optimization was applied to preserve the compact dimensions achieved by Stockmeyer while reducing wirelength to 32.50 units. This approach ensures that the overall floorplan boundary is maintained, providing a **dimensionally consistent layout** suitable for practical chip integration.

Comparison and Analysis: - The Stockmeyer algorithm significantly reduces area by efficiently arranging blocks and minimizing dead spaces. It provides a solid foundation for subsequent optimization stages. - Unconstrained HPWL optimization achieves the **lowest wirelength**, improving signal integrity and reducing potential delays; however, it may slightly alter the floorplan geometry. - Simulated Annealing explores global minima and balances wirelength with layout uniformity, preventing excessive distortions but slightly increasing HPWL compared to unconstrained HPWL. - Constrained HPWL optimization preserves the Stockmeyer floorplan's compact dimensions while achieving a moderate wirelength reduction.

This stage ensures **geometric consistency**, making it suitable for hierarchical or pre-sized chip designs.

Conclusion: The combined optimization process demonstrates a **stepwise improvement** in both area and wirelength efficiency. Stockmeyer provides a compact and area-efficient layout. Subsequent HPWL and SA optimization reduce interconnect costs and improve routing performance. The constrained HPWL approach strikes the **optimal balance** between performance and structural stability, maintaining the compactness of the Stockmeyer layout while reducing wirelength effectively. Overall, the pipeline ensures a **dimensionally stable, area-efficient, and wirelength-optimized floorplan**, which is critical for practical VLSI chip implementation.

F. Discussion

The results demonstrate that:

- The **Stockmeyer algorithm** efficiently minimizes the floorplan area while maintaining a structured slicing layout.
- The **unconstrained HPWL optimization** reduces interconnect length effectively but may alter the original floorplan dimensions.
- **Simulated Annealing** offers further wirelength improvement and global optimization but allows flexible boundaries.
- The **constrained HPWL model** ensures that the overall geometric boundary remains unchanged, preserving design compactness while providing balanced wirelength improvement.

Overall, these results highlight the trade-off between geometric consistency and interconnect minimization. While unconstrained methods achieve superior HPWL reduction, constrained optimization provides a more physically realizable layout suitable for hierarchical and fixed-outline floorplanning in VLSI systems.

TABLE I
AREA OPTIMIZATION USING STOCKMEYER ALGORITHM

Floorplan Stage	Width (W)	Height (H)	Area (A = W×H)
Initial Floorplan	15	12	180
Stockmeyer Optimized	13	9	117

TABLE II
WIRELENGTH (HPWL) AND TOTAL COST COMPARISON ACROSS OPTIMIZATION STAGES

Optimization Stage	Wirelength (HPWL)	Total Cost (Area + HPWL)
Initial Floorplan	(not provided)	(Area + HPWL unknown)
Stockmeyer Floorplan	37.00	117 + 37.00 = 154.00
Stockmeyer + HPWL (Unconstrained)	31.00	117 + 31.00 = 148.00
Stockmeyer + Simulated Annealing (Unconstrained)	33.50	117 + 33.50 = 150.50
Constrained HPWL Optimization	32.50	117 + 32.50 = 149.50

G. Limitations

While the current study demonstrates the effectiveness of Stockmeyer-based area optimization combined with HPWL and Simulated Annealing for wirelength minimization, there are several limitations that could be addressed in future work:

- The number of blocks and nets considered in this study is relatively small, which may not fully capture the complexity of large-scale VLSI designs.
- The analysis does not account for routing congestion, buffer insertion, or timing-driven constraints, which are critical for practical chip design.
- Thermal effects and power consumption are not incorporated, which may impact the feasibility of the floorplan in high-performance or multi-core designs.
- Process variations and parasitic effects are ignored, which can affect the accuracy of the wirelength and area estimation.
- Explore **hierarchical floorplanning** to optimize larger designs by partitioning the chip into sub-blocks for better scalability.
- Implement **machine learning-based prediction models** to estimate optimal block placements and wirelengths, reducing computation time for large designs.

H. Future Work

Based on the limitations identified above, several directions can be pursued to enhance the floorplanning methodology and improve overall chip performance:

- Incorporate **timing-driven optimization** to minimize critical path delays in addition to area and wirelength.
- Extend the methodology to **multi-layer or 3D IC floorplanning** to better handle complex modern designs.
- Integrate **power-aware and thermal-aware optimization** to ensure efficient heat dissipation and reduced power consumption.
- Apply **advanced metaheuristic algorithms**, such as Genetic Algorithms or Particle Swarm Optimization, to explore more global solutions.
- Perform a **sensitivity analysis** to evaluate the impact of variations in block sizes, netlist configurations, and Polish expressions on final area and wirelength.
- Incorporate **routing congestion estimation** to improve the practicality of the floorplan in real-world VLSI designs.

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